EMPLOYEE WORK ANALYSIS – DATA CLEANING DOCUMENTATION

Introduction:

The objective of this project is to transform a raw employee dataset into a clean, structured, and analysis-ready format. The raw dataset contained multiple inconsistencies such as missing values, incorrect formats, redundant columns, and invalid entries. These issues would directly affect the accuracy of any analysis or dashboard built on top of it.

The cleaned dataset was prepared using systematic data preprocessing techniques in Excel. The goal was to improve data quality, ensure consistency, and make the dataset suitable for further analysis such as performance evaluation, salary insights, and employee distribution.

Raw Data Issues Identified:

After examining the raw dataset, the following major problems were identified:

Missing Values

- Employee names were missing in several rows.
- Department column had multiple null values.
- Salary values were missing in some entries.
- Performance ratings were incomplete.
- Gender column had a high number of missing values.

- Date of Joining had missing records.
- This is a serious issue because incomplete data leads to biased or incorrect analysis.

Inconsistent Column Structure:

- Column name **"Name"** was not standardized.
- "Performance Rating" had inconsistent formatting.
- Extra column **"Country"** was present but not useful for analysis.
- Keeping irrelevant columns reduces clarity and increases noise.

Invalid and Incorrect Entries:

- Email column contained invalid formats like invalid _ email.
- Gender column had inconsistent values such as:
  - "Male"
  - "male"
  - "Other"
  - Missing values

This creates grouping issues in dashboards and analysis.

Data Formatting Issues:

- Text data was not standardized (mixed case, extra spaces).
- Dates were not consistently formatted.
- Column naming was not clean for professional reporting.

Data Cleaning Steps Performed:

This is the most important section — what you actually did.

Column Renaming

Renamed "Name" → **"Employee Name"**

Removal of Unnecessary Columns

Removed the **"Country"** column.

Handling Missing Values:

Employee Name:

Missing names were filled with appropriate values (e.g., standard placeholder or corrected entries).

Department:

Null values were replaced using logical assumptions or consistent department assignment.

Salary:

Missing salary values were handled using aggregation (average or estimated values).

Performance Rating:

Missing ratings were filled using consistent values or derived logic.

Gender:

Missing values were replaced with standardized categories such as:

- Male
- Female
- Other
- Unknown

Standardization of Text Data:

Applied functions like:

- TRIM → removed extra spaces
- CLEAN → removed unwanted characters
- PROPER/LOWER → standardized text format

Email Validation and Correction:

Invalid email entries were corrected into a proper format.

Data Type and Format Correction:

Ensured:

- Salary is numeric
- Dates are in proper date format
- Ratings are consistent

Final Null Value Reduction:

Before cleaning:

- Multiple columns had significant null values (e.g., Gender had 44 missing entries)

After cleaning:

- Almost all columns have **zero missing values** (except minimal unavoidable cases like dates)

Cleaned Data Overview:

After cleaning, the dataset has the following improvements:

- Total Rows: **150 (unchanged)**
- Columns reduced from **9 → 8**
- All major missing values handled
- Standardized and consistent format across all columns

Final Columns

- Employee ID
- Employee Name
- Department

- Salary
- Date of Joining
- Performance Rating
- Gender
- Email

Conclusion:

The raw dataset was not suitable for analysis due to missing values, inconsistencies, and poor formatting. Through systematic data cleaning, the dataset was transformed into a structured and reliable format.